Machine Learning Based Mini Project

Moisture Estimation of Power Transformer insulation by Frequency Domain Dielectric Spectroscopy (FDS) Measurements

Learning Outcome – *(LO4) – Describe Non-destructive tests used in high voltage engineering and demonstrate their applications on high voltage apparatus*

Background

Power transformers are among the most critical and expensive assets in electrical power transmission and distribution systems. The reliability and lifetime of a transformer are highly dependent on the condition of its insulation system, which mainly consists of mineral oil, cellulose paper, and pressboard insulation. During long-term operation, thermal, electrical, chemical, and environmental stresses lead to insulation aging and gradual moisture accumulation within the cellulose insulation. Excessive moisture in transformer pressboard significantly reduces dielectric strength, accelerates insulation degradation, increases dielectric losses, and may eventually result in catastrophic transformer failure. Therefore, accurate assessment of transformer insulation moisture content is an important aspect of transformer condition monitoring and asset management.

Among the available diagnostic techniques, Frequency Domain Dielectric Spectroscopy (FDS) has become a widely accepted non-destructive method for evaluating the condition of transformer insulation systems. In FDS measurements, dielectric response characteristics of the insulation are measured over a wide frequency range, typically from 1000 kHz down to 0.1 mHz, usually with three measurement points per decade. The most important dielectric parameter obtained from FDS measurements is the dielectric dissipation factor or loss tangent ($\tan \delta$), which is highly sensitive to insulation condition.

The measured FDS response is influenced by several parameters including:

Moisture content (MC) of the pressboard insulation,
Insulation temperature,
Conductivity of transformer oil, and
Geometric insulation parameters represented by X and Y parameters.

By varying these parameters, dielectric response curves with different characteristics can be generated and used for insulation condition assessment. In practical applications, interpretation of FDS measurements becomes challenging because of the large amount of data obtained over a wide frequency spectrum and the complex nonlinear relationship between dielectric response and insulation condition.

Machine Learning (ML) techniques provide an effective approach for extracting meaningful information from large FDS datasets and automating transformer condition assessment. In particular, linear dimensionality reduction methods such as Principal Component Analysis (PCA) can be used to reduce the high-dimensional FDS dataset into a smaller set of significant features while preserving the most important information. These reduced features can then be used with classification algorithms such as Support Vector Machines (SVM) to classify insulation condition accurately.

For this project, a dataset consisting of FDS measurements from prepared about 70 different transformer pressboard insulation samples will be provided. The dataset has been generated considering:

Six different pressboard moisture contents (0.3%, 0.8%, 1.6%, 2.4%, 3.8%, 5.2%)
Four different temperatures (35°C, 45°C, 55°C, 65°C)
Three different oil conductivity levels (σ), "low", "medium" and "high" and
Fixed geometric parameters (X and Y).

Students are expected to analyse the FDS data, perform feature extraction and dimensionality reduction (i.e. using PCA or any other method), and develop classification model (i.e. using SVM or any other method) for transformer insulation condition assessment.

The insulation condition categories are defined according to the moisture content (MC) of the transformer pressboard as follows:

Dry insulation: MC < 1%
Intermediate insulation: 1% ≤ MC ≤ 3%
Wet insulation: MC > 3%

This mini project introduces the students to the application of Artificial Intelligence (AI) and Machine Learning techniques in High Voltage Engineering and transformer diagnostics.

More details related to the modelling and machine learning approach can be found in the following publication:

Tharaka Wijethunga, Pawara Tharkana, Amila Wimalaweera, Janaka Wijayakulasooriya, Sarath Kumara, Kapila Bandara and Manjula Fernando, "A Machine Learning Approach for FDS Based Power Transformer Moisture Estimation" IEEE Conference on Electrical Insulation and Dielectric Phenomena (CEIDP), Vancouver Canada, December 2021, DOI: **10.1109/CEIDP50766.2021.9705366**.

Objectives

To understand the influence of insulation parameters on FDS characteristics,
To investigate the challenges associated with direct interpretation of FDS measurements,
To apply Machine Learning techniques for transformer insulation condition assessment,
To perform dimensionality reduction methods such as Principal Component Analysis (PCA) for improvement (optional),
To develop a classification model using Support Vector Machine (SVM) or other suitable ML or NN based method
To evaluate the performance of the developed ML model

The following tasks shall be completed using the provided FDS dataset consisting of transformer pressboard insulation samples.

Assignment

Task 1 – Analysis of FDS Characteristics

Students shall first investigate how different insulation parameters affect the dielectric loss tangent (tan δ) characteristics of transformer pressboard insulation.

(a) Effect of Moisture Content on FDS Characteristics - Plot the FDS loss tangent curves for comparing Dry insulation: MC = 0.3% and Wet insulation: MC = 5.2% (for a selected oil conductivity i.e. low, and temperature, T = 35°C)

(b) Effect of Temperature on FDS Characteristics - Plot FDS curves comparing T = 35°C and T = 65°C (For a selected MC i.e. 5.2% and oil conductivity, i.e. low)

(c) Effect of Oil Conductivity on FDS Characteristics - Plot FDS curves comparing a low oil conductivity and a high oil conductivity (For a selected MC i.e. 0.3% and temperature T = 35°C)

(d) Discussion on Direct Moisture Estimation Using FDS

Based on the above observations, students shall comment on the difficulties associated with directly estimating transformer pressboard moisture content from raw FDS measurements.

Task 2 – Principal Component Analysis (PCA) – (Optional)

Since FDS measurements contain a large number of frequency-domain features i.e 22 features, students shall apply Principal Component Analysis (PCA) for dimensionality reduction and propose suitable number of PCs required to represent the dataset adequately.

Task 3 – Machine Learning Based Classification

Students shall develop a Machine Learning model to classify transformer insulation condition into, Dry insulation, Intermediate insulation and Wet insulation. The Support Vector Machine (SVM) algorithm is recommended or may use any relevant ML algorithm.

Task 4 – Performance Evaluation

Students shall evaluate the developed ML model using suitable performance metrics such as Confusion Matrix or any suitable method. If the evaluated performances are not satisfactory, Students may explore other possibilities.

Task 5 – Further analyses – (Optional)

Students may explore possibilities of further improvements (if necessary) with AI based techniques

Reporting

Submit a short report consisting of the results and discussions part including (i) FDS characteristics affected by moisture content, temperature, oil conductivity (ii) Training and testing by ML based algorithm and (iii) Suggestions for further improvement.

NB: AI-based tools including "ChatGPT" may be used for improving presentation quality and assisting conceptual understanding, but students must not directly reproduce AI-generated content as their own work.

Appendix (Selected FDS results)

Loss Tangent	MC = 0.3 %				MC = 5.2 %			
	Low σ		High σ		Low σ		High σ	
Frequency [Hz]	t=35°C	t=65°C	t=35°C	t=65°C	t=35°C	t=65°C	t=35°C	t=65°C
0.0001	0.313096	2.091101	0.307918	2.046144	36.21963	23.02545	102.0419	32.75299
0.000215	0.263748	1.334595	0.25406	1.2926	32.9442	19.07925	93.20651	26.97254
0.000464	0.216453	0.896971	0.197944	0.847249	31.34438	15.94536	89.2703	22.45056
0.001	0.188339	0.65312	0.151697	0.58094	30.86735	14.48841	88.72255	20.36095
0.00254	0.194921	0.539454	0.120486	0.419629	30.85915	14.21593	89.93984	19.95693
0.00464	0.264037	0.533855	0.11108	0.318495	30.17306	14.91378	89.84173	20.88978
0.01	0.42875	0.662066	0.115364	0.256894	27.15297	16.55695	83.5404	23.11502
0.0215	0.761844	1.004713	0.12673	0.228164	21.12753	19.33751	67.93073	26.89636
0.0464	1.390709	1.683174	0.148983	0.235539	14.05338	23.08206	47.50836	31.99134
0.1	2.31657	2.59952	0.202625	0.293298	8.297283	26.17424	29.63726	36.16944
0.215	2.726118	2.828544	0.342646	0.44101	4.688125	26.24372	18.04831	36.18077
0.464	1.931308	1.935255	0.66658	0.765828	2.728814	21.54431	11.71629	29.65002
1	1.013975	1.014529	1.3312	1.423919	1.664749	14.79301	8.028823	20.31382
2.15	0.485241	0.487681	2.314646	2.388106	1.02527	8.7735	5.330397	12.00266
4.64	0.227509	0.229424	2.668452	2.691952	0.623351	4.919232	3.252753	6.691988
10	0.106471	0.107442	1.826735	1.820588	0.375748	2.756162	1.819552	3.704557
21.5	0.050127	0.050572	0.944903	0.938059	0.226816	1.596693	0.962208	2.103867
46.4	0.024008	0.024083	0.450089	0.446468	0.138914	0.953163	0.497414	1.221625
100	0.011972	0.011869	0.211406	0.209521	0.086853	0.577117	0.258648	0.716084
215	0.006389	0.00619	0.098799	0.097762	0.055443	0.349311	0.136461	0.418884
464	0.003843	0.00359	0.046342	0.045698	0.036327	0.213266	0.074054	0.247042
1000	0.002761	0.002466	0.022775	0.022294	0.024978	0.134161	0.042898	0.150666